Do Now · retrieval

Lesson 057 · C# Data Structures Mini-project: Library Loans

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does a constructor initialise?

Q2

What does Queue<T>.Dequeue do?

Q3

Why choose a unique ID for a record?

Q4

A zero-indexed C# int[,] array a has 3 rows and 2 columns. Nested loops visit r=0 to 2 and c=0 to 1 inclusive, setting a[r,c]=10*r+c. Determine the final values in the last row, from left to right.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Class Loan models a single loan request, containing a student and book title initialised in its constructor.

Core idea 2

Model — Queue<Loan> stores objects in requested service order. Enqueue adds a new request and Dequeue retrieves oldest.

Core idea 3

Evaluate — For a more complete system, use a Dictionary keyed by unique loan ID, validation and tests; do not treat a name as necessarily unique. A queue exists only while the program keeps it in memory; a text file can persist loan records. In OCR examination pseudocode, openWrite plus writeLine writes a file (overwriting existing contents), while openRead plus readLine retrieves lines. Always close() the file.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

model — Representation of a real-world concept.

constructor — Initialises new object.

Queue<T> — Generic FIFO collection.

validation — Checks input meets rules.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace two loan objects in a queue: Ida first, Tom second. Which is processed first?

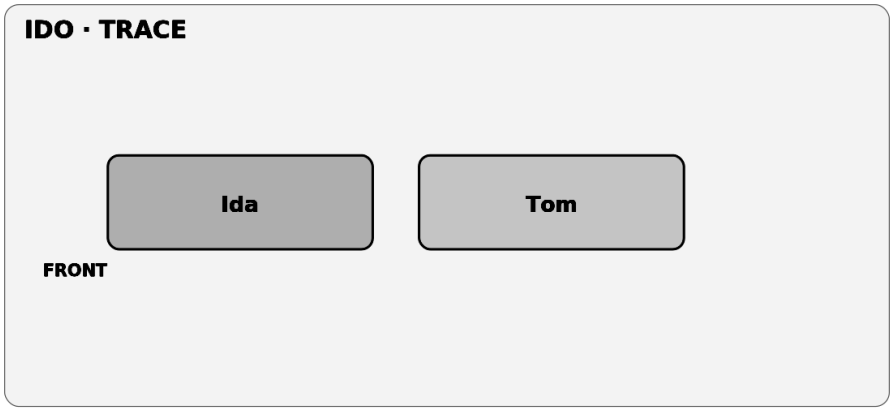
Follow the worked method

1. Explain — Queue is FIFO, so the first enqueued request is processed first.

2. Trace the file contents — First writeLine stores Ada,Holes , then the second stores Ben,Dune . They become two separate text lines. This is a simple illustrative format, not a robust CSV parser.

3. Explain the effect — f.close() finishes the write operation and releases the file. Do not confuse storing records in a queue or dictionary with saving them permanently.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain
A dictionary retrieves by key; it does not enforce FIFO service order.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain two reasons Loan is a class rather than separate parallel arrays for pupil and book. [4]

Question 2

Write C# creating a Queue<Loan>, enqueue two instances, then safely dequeue one using Count. [8]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Outline how to prevent duplicate loan IDs and reject empty student names; then name the OCR file operations for writing one saved loan and reading a line back. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A pupil's name is always guaranteed unique, so it is a perfect database key.

Use key vocabulary from memory

1. Explain model in your own words.

2. Explain constructor in your own words.

3. Explain Queue<T> in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.